

LAIZHEN LI

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EDUCATION

University of Chinese Academy of Sciences, China
Master of Engineering in Computer Technology
Advisor: Xitong Gao

Sep. 2024–Present (Expected Jul. 2027)
GPA: 3.71/4.0

Henan University of Engineering, China
Bachelor of Engineering in Software Engineering

Sep. 2019–Jul. 2023
Average: 81/100

RESEARCH INTERESTS

I work on trustworthy and efficient AI, with a current focus on LLM safety, agent security, and agent efficiency.

RESEARCH EXPERIENCE

Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences

Jun. 2025–Present

Master's Student

Advisor: Xitong Gao

LLM Safety — StreamGuard

- Designed and implemented **StreamGuard**, a lightweight, plug-and-play decoding-time defense against jailbreak attacks that self-reviews partial generations and inserts safety prompts when unsafe content is detected, without fine-tuning.
- Evaluated StreamGuard on Llama-3.1-8B-Instruct and Qwen2.5-7B-Instruct against AutoDAN, GCG, and DeepInception; achieved a mean ASR of **1.7%**, compared with **20.3%** for the best-performing defense baseline and **60.3%** without defense, across six model–attack settings.

Agent Security — A2M

- Designed and implemented **A2M (Attraction to Manipulation)**, a two-stage black-box attack that optimizes tool meta-data (tool names and descriptions) to increase selection likelihood, then refines return payloads using execution traces to manipulate downstream reasoning.
- Evaluated on LiveMCPBench across five frontier models; when optimized directly against GLM-4.6, A2M achieved a mean ASR of **74.4%** (64.3–92.9%) across three ASR-based objectives (information exfiltration, unauthorized environment modification, and reasoning derailment) and increased token consumption by **32.4×** on a fourth, token-wasting objective; the optimized attacks remained effective when transferred to four unseen models.

Agent Efficiency — Growing Harness

- Designed and implemented **Growing Harness**, a failure-guided framework that starts from a strategy-free scaffold and incrementally grows it into an executable agent harness through function-level execution traces and localized code edits, shifting recurring control logic from LLM inference to reusable program code.
- Evaluated on BrowseComp-Plus and WebArena-Verified across three backbone models; achieved a mean task success rate of **42.2%** across six benchmark–model settings, versus **23.6%** for the ReAct-style Tool-Calling baseline (**+18.6 pp**), while reducing LLM calls by **76.0–91.8%** and inference cost by **74.4–98.6%**.

PUBLICATIONS

[1] A2M: Trace-Optimized Agent Hijacking in the MCP Ecosystem
Laizhen Li, Xuan Wang, Peicheng Zhao, Juanjuan Zhao, Kejiang Ye, Cheng-zhong Xu, and Xitong Gao.
AACL-IJCNLP 2026, accepted.

[2] StreamGuard: A Streaming-based Defense Against Jailbreaking Attacks in Large Language Models
Laizhen Li, Xitong Gao, Xuan Wang, Juanjuan Zhao, and Kejiang Ye.
ICONIP 2025. [Paper]

[3] JADE: Jailbreak-Aware Dynamic Defense Enhancer for LLMs via Reinforcement Learning
Peicheng Zhao, Laizhen Li, Xitong Gao, Juanjuan Zhao, and Kejiang Ye.
IEEE BigData 2025. [Paper]

MANUSCRIPTS UNDER REVIEW

[1] Grow the Harness, Not the Context: From Strategy-Free Scaffolds to Reusable Specialist Agents
Laizhen Li, Jiarui Li, Juanjuan Zhao, Kejiang Ye, Ye Li, Cheng-zhong Xu, and Xitong Gao.
AAAI 2027. Under review.

[2] Co-Evolving Poison and Defense Harnesses: Adversarial Arms Races for Indirect Prompt Injection in Tool-Integrated Agents
Peicheng Zhao, Laizhen Li, Juanjuan Zhao, Kejiang Ye, and Xitong Gao.
AAAI 2027. Under review.

SKILLS

Programming & Tools: Python, Bash, Linux, Git, Docker.

Machine Learning: PyTorch, Hugging Face Transformers, Weights & Biases, SFT, PPO, LoRA.

LLM Systems: OpenRLHF, vLLM, LangChain, MCP.